

# **FINAL REPORT**

## **Civil Engineering Insight Studio**

### **1. INTRODUCTION**

#### **1.1 Project Overview**

Civil Engineering Insight Studio is a web-based Generative AI application designed to automate structural analysis and documentation from construction site images. The system leverages Google Gemini Vision API to analyze images and generate structured engineering reports.

The application allows users to:

- Upload construction site images
- Enter structural analysis request
- Generate structured engineering report
- Identify materials and structural components
- View formatted AI-generated output

#### **1.2 Purpose**

- The purpose of this project is to reduce manual documentation effort in civil engineering projects, provide AI-assisted structural analysis, and demonstrate real-world integration of multimodal Generative AI within a deployable web application

### **2. IDEATION PHASE**

#### **2.1 Problem Statement**

Civil engineers and construction supervisors spend significant time manually analyzing construction images and preparing structured reports. Traditional documentation methods are time-consuming and subjective.

| Customer Problem Statement                                                   |                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Context: Civil Engineering Insight Studio – AI Powered Analysis Tool |                                                                                                                                                                                                                                                |
| I am<br><br>Describe the customer and their attributes                       | <div>PS-1: I am a civil engineer assessing and documenting infrastructure projects</div> <div>PS-2: I am a construction site supervisor overseeing building and bridge construction</div>                                                      |
| I'm trying to<br><br>List the thing they are trying to achieve here          | <div>PS-1: I'm trying to analyze construction site images and produce detailed engineering reports</div> <div>PS-2: I'm trying to monitor progress and identify materials and structural components</div>                                      |
| but<br><br>Describe the problems or barriers that get in                     | <div>PS-1: But I find manually describing materials, components, and construction stages consumes too much time and is subjective</div> <div>PS-2: But tracking project progress without automated tools lacks precision and consistency</div> |
| which makes me feel<br><br>Describe the emotions the result from experience  | <div>PS-1: Which makes me feel overwhelmed and prone to human error</div> <div>PS-2: Which makes me feel frustrated, stressed, and inefficient in managing site documentation</div>                                                            |

2.2 Empathy Map Canvas

Target User: Civil Engineer / Site Supervisor

Pain Points:

- Time-consuming manual analysis
- Risk of missing structural details
- Inconsistent reporting format
- Heavy workload under tight deadlines

2.3 Brainstorming

Among multiple AI-based ideas, the AI-powered civil structure analysis system was selected due to its strong industry relevance, technical feasibility, and clear problem–solution alignment.



## Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

 10 minutes to prepare  
 1 hour to collaborate  
 2-8 people recommended



### Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

 10 minutes

---

A

**Team gathering**  
 Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B

**Set the goal**  
 Think about the problem you'll be focusing on solving in the brainstorming session.

C

**Learn how to use the facilitation tools**  
 Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) →

1

### Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

 5 minutes

---

PROBLEM

How might we [your problem statement]?



### Key rules of brainstorming

To run a smooth and productive session

 Stay in topic.

 Encourage wild ideas.

 Defer judgment.

 Listen to others.

 Go for volume.

 If possible, be visual.

### 3. REQUIREMENT ANALYSIS

#### 3.1 Functional Requirements

- Accept construction image upload
- Accept structural analysis request
- Generate structured report via Gemini API
- Display formatted engineering output
- Handle API errors gracefully

#### 3.2 Non-Functional Requirements

- Usability – Simple Streamlit interface
- Performance – Response within 3–10 seconds
- Reliability – No crash during API failure
- Security – API key stored securely using environment variables
- Scalability – Cloud deployable stateless architecture

### 3.3 Technology Stack

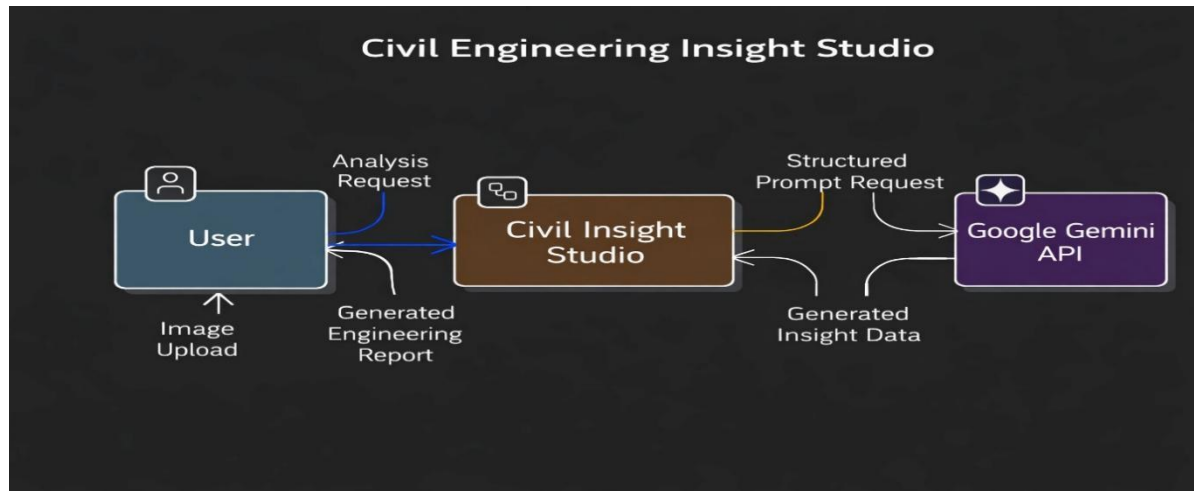
Frontend: Streamlit

Backend: Python

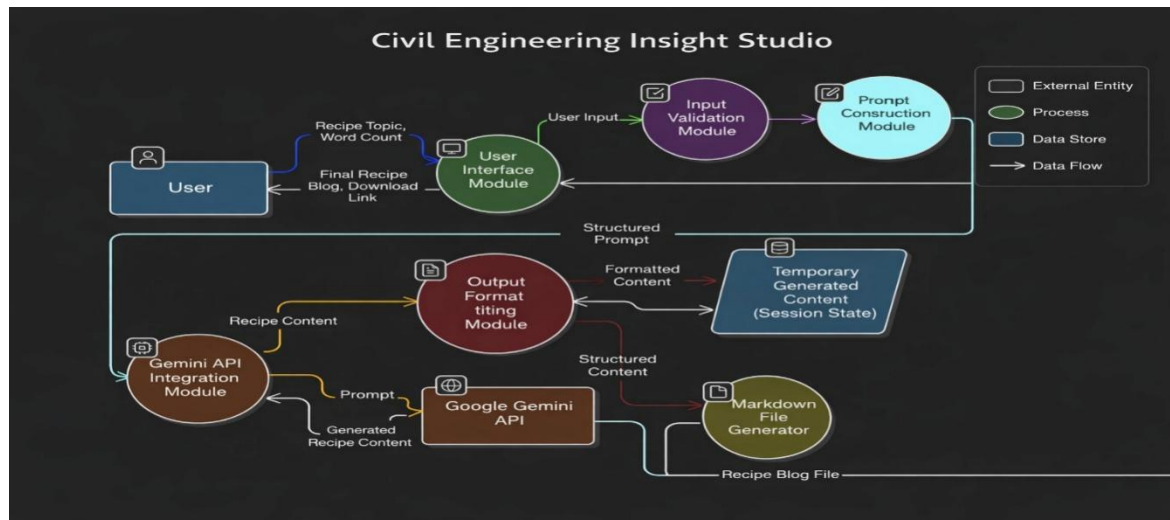
AI Model: Google Gemini Vision API

Deployment: Local / Streamlit Cloud

DFD Level 0 (Industry Standard)



DFD Level 1 (Industry Standard)



The system follows a stateless architecture with no persistent database.

## 4. PROJECT DESIGN

### 4.1 Problem–Solution Fit

Problem: Manual structural documentation is slow and inconsistent.

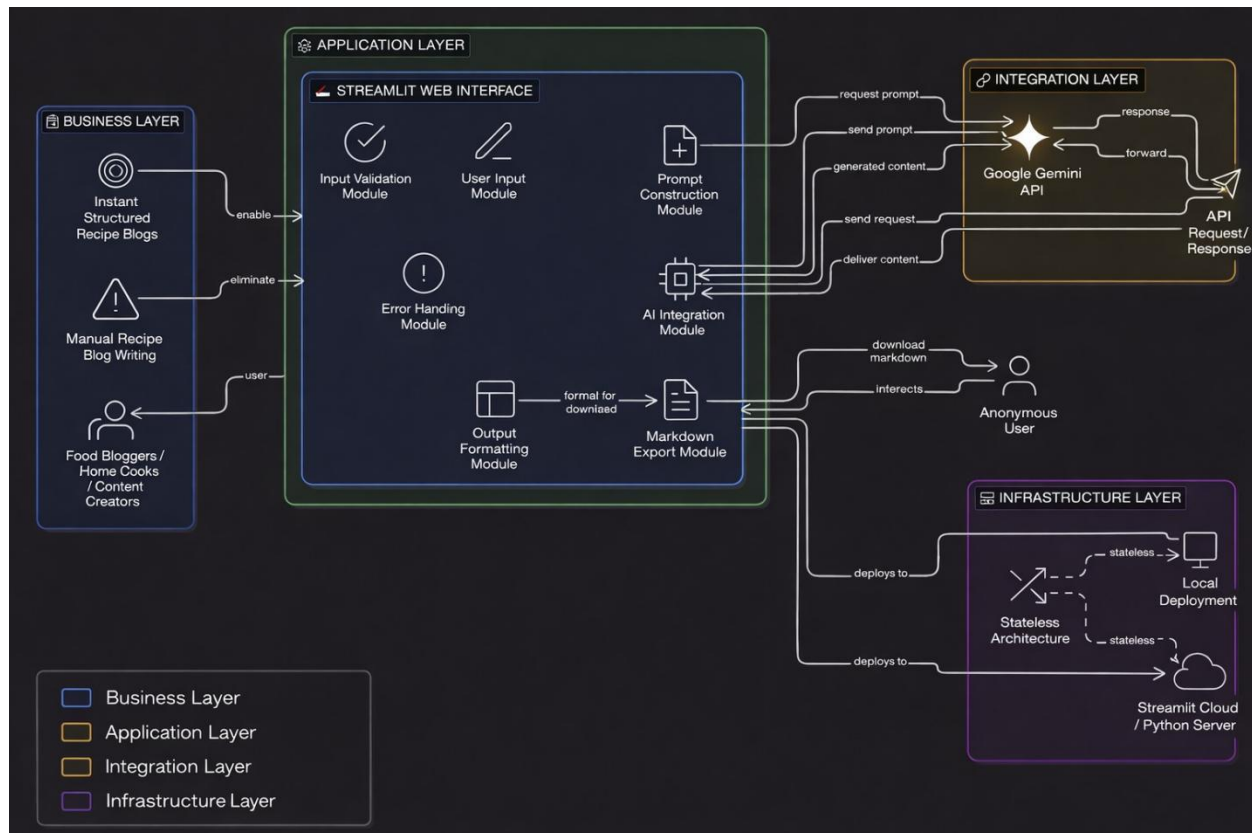
Solution: AI-powered image analysis generating structured engineering reports instantly.

|                                          |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                         |                                          |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Define CS, fit into CC                   | <b>1. CUSTOMER SEGMENT(S)</b><br>Who is your customer?<br>I.e. working parents of 0-5 y.o. kids                                                                                                                | <b>6. CUSTOMER CONSTRAINTS</b><br>What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.                                                                                                                                                                                   | <b>5. AVAILABLE SOLUTIONS</b><br>Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking          | Explore AS, differentiate                |
|                                          |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                         |                                          |
| Focus on J&P, tap into BE, understand RC | <b>2. JOBS-TO-BE-DONE / PROBLEMS</b><br>Which jobs-to-be-done (or problems) do you address for your customers?<br>There could be more than one; explore different sides.                                       | <b>9. PROBLEM ROOT CAUSE</b><br>What is the real reason that this problem exists?<br>What is the back story behind the need to do this job?<br>I.e. customers have to do it because of the change in regulations.                                                                                                                                                                                | <b>7. BEHAVIOUR</b><br>What does your customer do to address the problem and get the job done?<br>I.e. directly related: find the right solar panel installer, calculate usage and benefits;<br>Indirectly associated: customers spend free time on volunteering work (I.e. Greenpeace) | Focus on J&P, tap into BE, understand RC |
|                                          |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                         |                                          |
| Identify strong TR & EM                  | <b>3. TRIGGERS</b><br>What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.                                                | <b>10. YOUR SOLUTION</b><br>If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.<br>If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. | <b>8. CHANNELS of BEHAVIOUR</b><br><b>8.1 ONLINE</b><br>What kind of actions do customers take online? Extract online channels from #7                                                                                                                                                  | Extract online & offline CH of BE        |
|                                          | <b>4. EMOTIONS: BEFORE / AFTER</b><br>How do customers feel when they face a problem or a job and afterwards?<br>I.e. lost, insecure > confident, in control - use it in your communication strategy & design. |                                                                                                                                                                                                                                                                                                                                                                                                  | <b>8.2 OFFLINE</b><br>What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.                                                                                                                                           |                                          |

### 4.2 Solution Architecture

The system follows a layered architecture:

- Client Layer – Civil Engineer (Web User)
- Presentation Layer – Streamlit Interface
- Application Layer – Python Logic + Prompt Engineering
- Integration Layer – Google Gemini Vision API
- Infrastructure Layer – Local/Cloud Hosting



## 5. PROJECT PLANNING & SCHEDULING

Development was completed in two sprints covering UI development, Gemini API integration, structured report formatting, validation, and testing.

Average team velocity: 15–19 Story Points per sprint.

## 6. FUNCTIONAL AND PERFORMANCE TESTING

Functional tests validated image upload, API integration, and report generation.

Performance tests confirmed average response time within 3–10 seconds depending on API latency.

System remained stable during UAT.

## 7. RESULTS

The system successfully generated structured engineering reports including:

- Structure type identification
- Materials detection
- Structural components (beams, columns, slabs)
- Construction stage observations

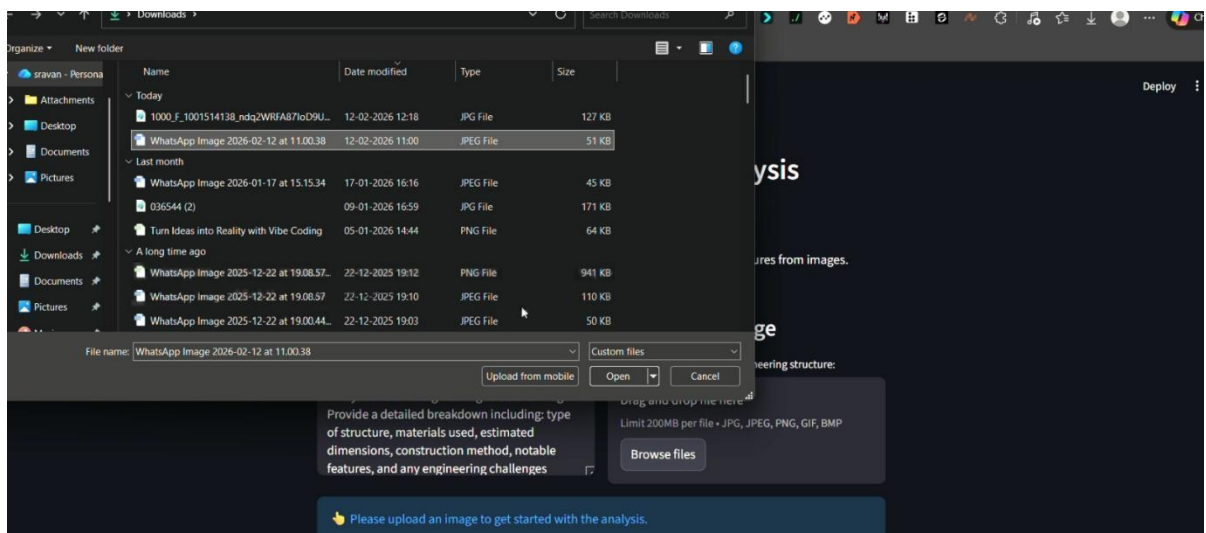
Outputs demonstrated consistent formatting and professional documentation quality.

## 7.1 Output Screenshots

- Main interface screen



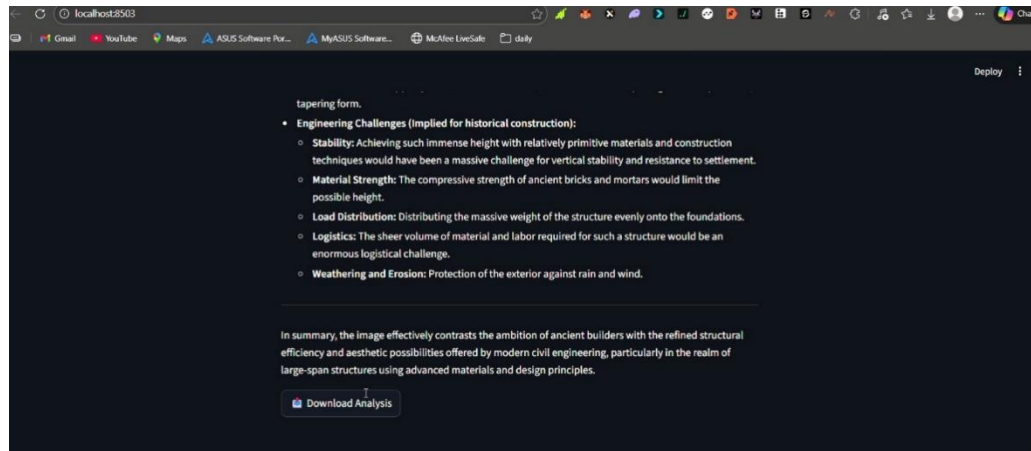
- Input section



- 

- 

- 
- A screenshot of a web browser window. The address bar shows "localhost:8060". The page displays a video player with a blurred image of a bridge. A "Downloads" panel is open on the right side of the browser, listing several files including "structure\_analysis.net", "1980\_F\_1981S141SE\_hsq2MHAATNDRULOODXGed", "DFAves.jpg", "ms\_python.python-32NLG.0.0.exe", "AddQuality.exe", "downloadsway", and "removed". At the bottom of the video player area, there is a red button labeled "Analyse Structure".



## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Significant time savings
- Automated structured documentation
- Industry-relevant AI integration
- Cloud deployable architecture

### Disadvantages

- Dependent on external AI API
- Requires internet connection
- API latency may vary

## 9. CONCLUSION

Civil Engineering Insight Studio demonstrates practical integration of multimodal Generative AI into civil engineering workflows. The system reduces manual documentation effort, improves reporting accuracy, and validates scalable AI-powered automation in the construction domain.

## 10. FUTURE SCOPE

- AI-based risk detection with bounding boxes
- Progress comparison between multiple images
- PDF report export feature
- User authentication & project history
- Cloud auto-scaling deployment

## 11. APPENDIX

### Source Code (app.py):

```
from dotenv import load_dotenv

import streamlit as st

import os

import google.generativeai as genai

from PIL import Image

import io


# Load environment variables from .env file

load_dotenv()


# Configure Google Generative AI with API key

api_key = os.getenv("GOOGLE_API_KEY")

if not api_key:

    st.error("Google API Key not found. Please add GOOGLE_API_KEY to your .env file.")

    st.stop()


genai.configure(api_key=api_key)
```

```
def get_gemini_response(input_text, image, prompt):
```

```
    """
```

Function to get response from Gemini model.

Args:

input\_text: User input text

image: List containing image data in the format required by Gemini

prompt: The prompt to send to the model

Returns:

str: The text response from the model

```
    """
```

```
    model = genai.GenerativeModel('gemini-2.5-flash')
```

```
    response = model.generate_content([input_text, image[0], prompt])
```

```
    return response.text
```

```
def input_image_setup(uploaded_file):
```

```
    """
```

Function to read the uploaded image and format it for Gemini Pro model.

Args:

uploaded\_file: The uploaded file object from Streamlit

Returns:

list: A list containing image data formatted for Gemini API

Raises:

FileNotFoundError: If no file is uploaded

"""

if uploaded\_file is not None:

# Read the file's binary data

bytes\_data = uploaded\_file.getvalue()

# Create image parts in the required format

image\_parts = [

{

"mime\_type": uploaded\_file.type,

"data": bytes\_data,

}

]

return image\_parts

else:

raise FileNotFoundError("No file uploaded")

def main():

"""Main function to run the Streamlit application."""

st.set\_page\_config(page\_title="Civil Engineering Structure Analysis",  
layout="centered")

st.header("🏠 Civil Engineering Structure Analysis")

st.subheader("Powered by Google Gemini AI")

```
# Description
```

```
st.markdown("""
```

```
This application uses Google's Gemini AI model to analyze civil engineering structures
```

```
from images. Upload an image of a structure and provide a prompt to get detailed analysis.
```

```
""")
```

```
# Create two columns for better layout
```

```
col1, col2 = st.columns([1, 1])
```

```
with col1:
```

```
st.subheader("📝 Input Prompt")
```

```
input_prompt = st.text_area(
```

```
"Enter your prompt for analysis:",
```

```
value="Analyze this civil engineering structure image. Provide a detailed breakdown including: type of structure, materials used, estimated dimensions, construction method, notable features, and any engineering challenges visible.",
```

```
height=150
```

```
)
```

```
with col2:
```

```
st.subheader("🖼️ Upload Image")
```

```
uploaded_file = st.file_uploader(
```

```
"Choose an image of a civil engineering structure:",
```

```
type=["jpg", "jpeg", "png", "gif", "bmp"]
```

```
)
```

if uploaded\_file is not None:

st.success("Image uploaded successfully!")

# Display the uploaded image

st.subheader("Uploaded Image Preview:")

image = Image.open(uploaded\_file)

st.image(image, use\_container\_width=True)

# Create a submit button

submit\_button = st.button("🔍 Analyze Structure", type="primary")

if submit\_button:

if input\_prompt.strip() == "":

st.error("Please enter a prompt for analysis.")

else:

with st.spinner("Analyzing the structure..."):

try:

# Prepare the image for Gemini

image\_data = input\_image\_setup(uploaded\_file)

# Get response from Gemini

response = get\_gemini\_response(

input\_text="Analyze the following image:",

image=image\_data,

prompt=input\_prompt

)

```

# Display the response

st.subheader("📊 Analysis Results:")

st.markdown(response)


# Add a copy button for convenience

st.download_button(

    label="📄 Download Analysis",
    data=response,
    file_name="structure_analysis.txt",
    mime="text/plain"
)


except FileNotFoundError as e:

    st.error(f"Error: {e}")

except Exception as e:

    st.error(f"An error occurred during analysis: {str(e)}")

else:

    st.info("👉 Please upload an image to get started with the analysis.")


if __name__ == "__main__":

    main()

```

## GitHub Repository

[SravanVeeramallu/Civil-Engineering-Insight-Studio-GEN-AI-project](https://github.com/SravanVeeramallu/Civil-Engineering-Insight-Studio-GEN-AI-project)

## Project Demo Link

[https://drive.google.com/file/d/1c85Dm538hVJlh2qba61Sga7iGm--3hNi/view?usp=drive\\_link](https://drive.google.com/file/d/1c85Dm538hVJlh2qba61Sga7iGm--3hNi/view?usp=drive_link)